

Surface and subsurface material selections influence the early outcomes of boreal upland forest restoration



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ABSTRACT

Soil is an essential component supporting the growth and maintenance of forests, providing anchorage, water, and nutrients for a diversity of plants. In Canada's boreal forest, surface soils differ widely in their chemical and physical conditions, ranging from coarse to fine textured mineral soils in the uplands to organic soils in the lowlands. Industrial disturbances in the boreal region require the selective salvage of surface and sub-soils from low- and upland areas during open pit mine operations, which can be used to reconstruct different soil profiles by arranging the materials in variable layers and thicknesses. The type of materials and their configuration in the reconstructed soil profile can have a profound effect on early forest establishment in reclaimed areas. We compared the growth of trembling aspen (*Populus tremuloides* Michx.), jack pine (*Pinus banksiana* Lamb.), and white spruce (*Picea glauca* Moench.) on sites that had different reconstructed soils caps using varying surface soil materials (salvaged lowland peat and upland forest floor material (FFM)), placement depths (10 or 30 cm for peat, 10 or 20 cm for FFM), and subsoil material types determined by salvage depth (Bm, BC, and C). Early seedling establishment and growth and soil and climatic parameters were monitored over a five-year period. Generally, seedling growth was greatest on FFM and appeared to be related to phosphorous availability, while peat as a surface soil reduced growth, likely due to delayed soil warming in the spring and cooler soil conditions that potentially limited resource availability. However, the greater water holding capacity of the organic matter in peat also provided a benefit for seedling growth that was apparent during water limiting climatic conditions. The underlying subsoil material also influenced growth, particularly later in establishment when roots occupied the deeper subsoils. Aspen growth was greatest when the subsoil was shallow salvaged and represented a weathered subsoil (Bm) compared to the more deeply salvaged, less weathered subsoils BC and C. Aspen and pine seedlings, with their larger roots systems, may have benefited from small increases in the silt fraction of the subsoils, likely due to the greater water holding capacity of these otherwise coarse textured sandy soils. Spruce regeneration responded marginally to soil treatments because of its overall slow growth-strategy and tolerance to resource limitations. Our results indicate that different soil capping strategies affect the performance of forest regeneration during post-mining site reconstruction, and boreal forest species may vary in responses according to their ecological life strategies.

1. Introduction

Above- and belowground structure, such as the layering of tree canopies and soil horizons, are essential elements in forest ecosystem functioning, creating conditions that drive forest diversity and biogeochemical and hydrological cycling between soils and plants (Hart and Chen, 2008; Macdonald and Fenniak, 2007). Anthropogenic disturbances, such as surface or open pit mining, often result in the disruption or loss of these relationships and functions (Macdonald et al., 2015; Pickell et al., 2013), since it requires the complete removal of the

existing vegetation and soil materials (topsoil, subsoil and geological overburden material) to access and extract the resource (Cumulative Environmental Management Association, 2009; Rowland et al., 2009). After mining and as part of the reclamation and restoration process, the topography, including soils and vegetation, of an entire landscape need to be reconstructed (Zipper et al., 2013). Given the scale of most surface mines, there are no natural disturbance analogues that could provide recovery trajectories to serve as a comparison; therefore, the reclamation and re-establishment of forests and their functional processes require novel approaches to assist in their recovery (Hiers et al., 2012;

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Hobbs et al., 2009; Jacobs et al., 2015).

Soil reconstruction is a major challenge in forest reclamation of upland areas because forest soils must sustain a diverse community of forest species over decades or centuries. Planting trees as a mixture that represent different successional stages can accelerate the development of a target forest community and reduce the risks associated with other re-vegetation methods like relying on natural colonization (Davis et al., 2012; Thompson et al., 2009). To successfully establish and grow on reclamation sites, planted tree seedlings require a sufficient supply of resources during the early establishment phase (Groninger et al., 2007); therefore, the materials used to reconstruct the soil cover must be able to support these demands and move these reclaimed areas towards fully functional forest ecosystems (Bussler et al., 2010; Howell et al., 2016). Although it is impossible to exactly recreate distinct soil horizons and their unique characteristics as found in natural soils, reclamation practices can attempt to emulate some of the same characteristics that provide short- and long-term ecosystem benefits (Burton and Macdonald, 2011; Naeth et al., 2012; Zipper et al., 2011, 2013).

The initial step in forest reclamation is the reconstruction of geotechnically stable landforms using overburden (OB) material that was originally geologically situated above the resource and below the parent material of the surface soils (Toy and Chuse, 2005). The chemical and physical characteristics of the OB influence its reclamation capability, or conversely, its limitations in the mine closure landscape. The subsequent addition of a reconstructed soil cover, comprised of salvaged coversoil and subsoil materials, is a practical and effective strategy to mitigate potential limitations or environmental risks of unsuitable OB substrates (Macdonald et al., 2012; Zipper et al., 2013). Creating a suitable soil cover design with appropriate soil material horizons and thicknesses can help minimize the economic and ecological risk of failure in reclaimed areas (Cumulative Environmental Management Association, 2009; Macdonald et al., 2015). Once the landforms are built, salvaged subsoil materials are placed on top of the OB and covered with salvaged topsoil (coversoil) materials (Oil Sands Vegetation Reclamation Committee, 1998). The coversoil material and the nutrients and soil organic material (SOM) contained within play an important role in short-term nutrient cycling and water infiltration and storage (Berg and Laskowski, 2005; Zhuang et al., 2008). In contrast, the shallower subsoil horizons (e.g. weathered B-horizon) are typically characterized by lower SOM, microbial activity, and nutrient content than the topsoil material, but may have higher availability of less mobile nutrients, such as phosphorous, than the deeper parent material (C-horizon) (Ojekanmi and Chang, 2014; Wolken et al., 2010). Deeper C-horizon subsoils can, however, support the essential long-term availability of mineral nutrients through weathering, while also providing structural support for deep-rooted plants and long-term water storage (Smith et al., 2011; Strong and La Roi, 1985).

In the boreal forest region of northern Alberta, areas that are predominantly disturbed by open pit mining are dominated by lowland forests that have organic soils composed largely of peat rather than mineral soils (Solodzik et al., 1982). The use of salvaged upland forest floor material (i.e. organic L, F, H horizons and the upper mineral A horizon) is preferred for upland forest reclamation; however, due to its greater availability, peat salvaged from these lowland forests (i.e. treed bogs and fens) are commonly used as a soil cover in upland forest reclamation (Rowland et al., 2009; Ojekanmi and Chang, 2014; Pinno et al., 2012). The predominant use of peat can introduce different challenges in upland forest reclamation. These peat materials used as a reclamation coversoil can create wet conditions and cool temperatures in the root zone with their low bulk density and high organic content (Zhao and Si, 2019), which can negatively impact the growth of upland tree species that prefer warmer soil conditions (Landhäusser et al., 2001; Karst and Landhäusser, 2014) and seasonal fluctuations in water and nutrient availability during the growing season (Chapin III et al., 1986).

The Aurora Soil Capping Study (ASCS) was established in 2011 to

assess the impact of different reconstructed soil covers on early upland tree seedling and forest establishment at an operational scale. The objective of this study was to assess early establishment and growth of three boreal tree species (trembling aspen (*Populus tremuloides* Michx.), jack pine (*Pinus banksiana* Lamb.), and white spruce (*Picea glauca* Moench.)) that are commonly planted on reclaimed sites in northern Alberta in response to coversoil material type and its depth and the underlying subsoil material, and relate these responses to the soil physical (i.e. soil temperature, water content, texture) and chemical (i.e. nutrient availability, pH, electrical conductivity, sodium adsorption ratio) properties of the capping materials. We hypothesized that (i) fast growing, pioneer species (aspen and pine) would be more sensitive to soil cover materials than a slow growing, late-successional species (spruce); (ii) higher nutrient availability in the salvaged upland coversoil would support faster growth rates in all species than the salvaged lowland coversoil, while water availability would be higher in the organic-rich peat coversoil compared to the coarse-textured upland coversoil; and, (iii) that providing an additional layer of a selectively salvaged weathered subsoil material would provide additional benefits for tree growth compared to unweathered, deeply salvaged, and homogenized subsoil materials.

2. Methodology

2.1. Study site

The ASCS site, a large-scale (36 ha) reclamation experiment, is located in the Syncrude Aurora North-Mine lease, about 80 km north of Fort McMurray, Alberta, Canada (57°20'N, 111°31'W). The site is located within the central mixedwood natural subregion, which is characterized by a rolling terrain of upland and lowland forests. Mixed stands of trembling aspen and white spruce are often found growing on Luvisolic soils while pure jack pine stands are found on Brunisolic soils in the upland areas (Natural Regions Committee, 2006). Lowland bogs and fens are dominated by black spruce (*Picea mariana* Mill.) and tamarack (*Larix laricina* (Du Roi) K. Koch) stands that have developed on poorly drained organic soils (Natural Regions Committee, 2006). Growing season (May–September) climate normals (1981–2010) for the area had an average daily temperature of 13.4 °C and total precipitation of 284.3 mm (Government of Canada, 2019). The average daily temperatures and total growing season precipitation during the first five growing seasons is provided in Table A1.

2.2. Experimental design

The reclamation site is on a plateau with a gentle slope (< 5%) and an east – west aspect at an elevation of 350 m on an OB dump. The dump was constructed using lean oil sand (LOS) OB material which has a sandy loam texture, neutral pH, and low bitumen concentration (average 2.7%). The grading of the OB on the ASCS surface landform was completed in 2011 and soil materials were placed prior to the spring of 2012. Thirteen different soil cover treatments were randomly assigned across the study site, and each soil treatment was structured into 1-ha cells and replicated three times within the split-plot factorial design, with soil treatment as the plot effect and tree planting treatment as the split effect (Fig. A1). For this study, we used six of the soil treatments; they varied in the type and thickness of coversoil material placed over three differing subsoil layers (Fig. 1).

Two coversoil materials were used in this study. Peat coversoil was salvaged to mineral soil contact (approximately 3–4 m) from a lowland black spruce and tamarack-dominated forest. The second coversoil was a forest floor material (FFM) that was salvaged to a maximum depth of 15 cm in a jack pine-dominated upland forest, which was underlain by a coarse-textured (predominately loamy sand), Brunisolic soil. At that salvage depth the organic litter layers (L,F,H) and the underlying mineral A horizon and potentially a portion of the B horizon were

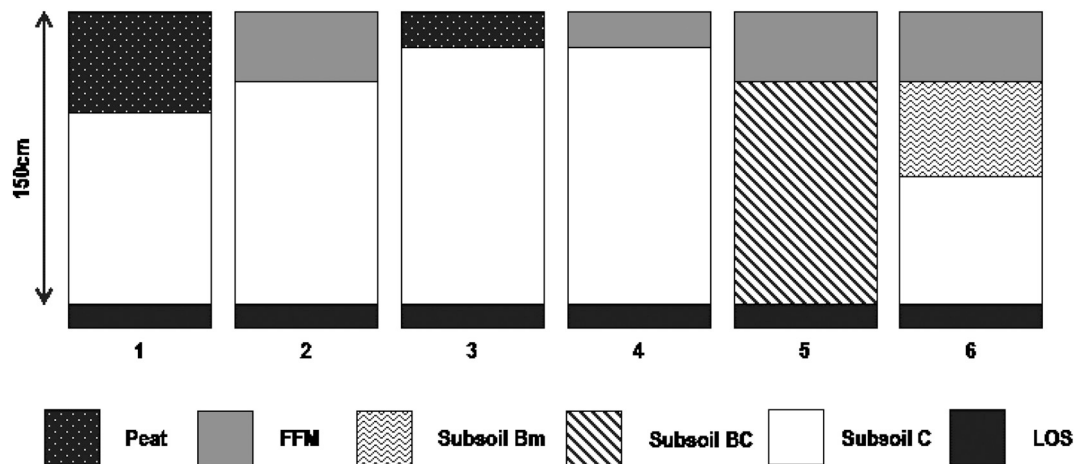


Fig. 1. Six of the thirteen soil layering treatments at the Aurora Soil Capping Study (ASCS) are shown. Treatment 1 is 30 cm of Peat over 120 cm of Subsoil C; treatment 2 is 20 cm of FFM over 130 cm of Subsoil C; treatment 3 is 10 cm of Peat over 140 cm of Subsoil C; treatment 4 is 10 cm of forest floor material (FFM) over 140 cm of Subsoil C; treatment 5 is 20 cm of FFM over 100 cm of Subsoil C, and; treatment 6 is 20 cm of FFM over 30 cm of Subsoil Bm over 100 cm of Subsoil C. All treatments were placed over lean oil sands (LOS) overburden material. Physical and chemical characteristics of these materials are presented in Table 1.

included in the coversoil material. All coversoils were salvaged and directly placed on the reclamation site without storing the material in stockpiles. Three different subsoil materials (depending on salvage depth) were salvaged near the Brunisolic upland soils; therefore, all subsoils shared the same glaciofluvial geologic parent material. The Subsoil Bm was salvaged between a soil depth of 15 and 50 cm, the Subsoil BC material was salvaged between 50 and 100 cm, which included the B and underlying C horizons, and the Subsoil C material was salvaged from a depth of 15 to 250 cm, which included the Bm, B, and C horizons. The Bm and BC subsoil materials were salvaged three years prior to their placement and stockpiled while the subsoil C material was salvaged and directly placed. Depending on the assigned soil treatment, Peat coversoil was placed at a target thickness of 10 or 30 cm and the FFM coversoil was placed at 10 or 20 cm. Coversoils were underlain by different subsoil material types (i.e. Subsoils Bm, BC, or C) in a range of configurations. Layering the soil profile with a selectively salvaged Bm subsoil was designed to mimic a weathered subsoil layer diagnostic for natural Brunisolic soils. The other two subsoil treatments (BC and C) represent a more operational and economical approach where the B and C horizons are salvaged together and placed on site somewhat blended and homogenized. The total depth of the soil cover placed over the OB was 150 cm for all soil treatments used in this study (Fig. 1).

Within each soil treatment cell, four tree plots (25 m × 25 m) were established (Fig. A1). Three of the tree plots were planted with a single tree species of either trembling aspen, jack pine, or white spruce, while the fourth tree plot was planted with an even mixture of all three species. All seedling stock was grown from local seed sources and produced at a commercial nursery (Smoky Lake Forest Nursery Ltd.) under operational protocols described in Landhäusser et al. (2012). Aspen and spruce seedlings were sown into 615A StyroblockTM (Beaver Plastics Ltd., Acheson, Alberta), while pine seedlings were sown into 412A StyroblockTM (Beaver Plastics Ltd., Acheson, Alberta). All seedlings were planted as one-year-old seedlings (see Table A2 for initial seedling characteristics), where aspen seedlings averaged 30 cm tall, spruce 29 cm, and pine 18 cm at time of planting (Table A2). Seedlings were hand-planted in June 2012 at a regular 1 m × 1 m spacing (equivalent to 10,000 stems per hectare (sph)). The areas outside the tree plots in each soil treatment cell were planted with a mixture of the same tree species at a density of approximately 2000sph, as well as three native shrub species: pincherry (*Prunus pensylvanica* L.f.), green alder (*Alnus viridis* (Chaix) DC.), and saskatoon (*Amelanchier alnifolia* (Nutt.) Nutt. ex M. Roem.) at a density of about 800sph. For this study, we only present data from tree plots that were planted with a single species at 10,000sph.

Seedling growth from 2012 to 2016, total height of seedlings in 2016, and their relative growth (i.e. growth from 2012 to 2016 relative to initial 2012 height for among species comparisons) were compared in response to:

- (1) Coversoil material type (FFM and Peat) where soil cover treatments 1 and 2 were used (Peat30 (30 cm Peat over 120 cm Subsoil C) and FFM20 (20 cm FFM over 130 cm Subsoil C)) (Fig. 1)
- (2) Coversoil placement depth (soil cover treatments 1, 2, 3, and 4; i.e. Peat 10 (10 cm Peat over 140 cm Subsoil C); Peat30 (30 cm Peat over 120 cm Subsoil C); FFM10 (10 cm FFM over 140 cm Subsoil C); and FFM 20 (20 cm FFM over 130 cm Subsoil C)) (Fig. 1)
- (3) Different subsoil configurations with a 20 cm FFM coversoil (soil cover treatments 2, 5, and 6 were used (Subsoil C (20 cm FFM over 130 cm Subsoil C); Subsoil BC (20 cm FFM over 130 cm Subsoil BC); and Subsoil Bm (20 cm FFM over 30 cm Subsoil Bm over 100 cm Subsoil C)) (Fig. 1).

2.3. Measurements

Prior to the first growing season, samples of all soil material types were collected from each soil treatment cell and physical and chemical properties were analyzed. Soil texture was measured using the hydrometer method (Kroetsch and Wang, 2008). Soil pH, electrical conductivity (EC), and sodium absorption ratio (SAR) were measured using the saturated paste method as described by Miller and Curtin (2008). Plant available sulfate (SO_4^{2-}) was determined using the method described in Novozamsky et al. (2008) with inductively coupled plasma-optical emission spectroscopy. Ammonium (NH_4^+) was extracted using KCl and measured colorimetrically (Haney et al., 2006). Nitrate (NO_3^-), phosphate (PO_4^{2-}), and potassium (K^+) were extracted using the Modified Kelowna method; NO_3^- and PO_4^{2-} were measured colorimetrically while K^+ was measured with flame photometry (Ashworth and Mrazek, 2008). The mass of plant available nutrients per cubic meter was calculated using the nutrient concentrations (i.e. mg/kg), the bulk density of the soil material, and a set volume (i.e. 1 m by 1 m by capping material thickness). Total organic nitrogen (TON) and soil carbon were measured using the dry combustion methods described in Bremner (1996) and Nelson and Sommers (1996). Total organic carbon (TOC) was calculated as the difference between total carbon and inorganic carbon. The TOC% and a conversion factor of 2 were used to calculate the reported organic matter (OM) percentages (Nelson and Sommers, 1996). Physical and chemical properties of the different materials are summarized in Table 1.

Multiple monitoring systems were installed to record vadose zone water dynamics in all soil materials throughout 2013–2016. Two types of soil sensors were used to collect volumetric water content (VWC) and temperature data: 1) time domain reflectometry (TDR) sensors (Model 616, Campbell Scientific) were used to monitor in situ water content, and 2) thermal conductivity (TC) sensors (Model 229, Campbell Scientific) were used to monitor in situ temperatures. Sensors were installed 5 cm/15 cm (coversoil) and 35 cm/45 cm (subsoil) below the soil surface. Daily means from April through September were averaged to determine the average growing season soil temperature and VWC. Furthermore, the number of days with daily mean soil temperature above 5 °C were calculated from 2013 to 2016 because root growth of most boreal tree species can be severely restricted below soil temperatures of 5 °C (Chapin III, 1977; Karst and Landhäusser, 2014; Landhäusser et al., 2001).

A single square sub-plot (5 m × 5 m) was established in the center of each tree plot for the assessment of tree performance. Within the sub-plot, 16 seedlings were individually tagged and measured for heights (from ground to bud tip) and root collar diameter (RCD; at ground level) each August from 2012 to 2016. Additionally, two circular sub-plots with a radius of 2 m were established in the NW and SE corners of each tree plot, and trees within each circle were assessed for the same parameters as the center sub-plot. Tree plot averages were calculated from the three subplots to assess total seedling height in 2016 and their relative growth, while the average of tagged trees from the center sub-plot were used exclusively for the 2012–2016 annual growth rates.

To explore early growth allocation of seedlings in response to coversoil material, three seedlings in each tree plot for all three species were excavated in treatments 1 and 2 in 2014. To avoid disturbing the seedling measurement plots through the excavation, we identified seedlings outside the measurement plots that were representative of each treatment using the average RCD of the trees in the measurement plots. Seedlings were carefully excavated to capture most of the root system and cold stored in the field, and then later frozen in the laboratory until the final processing. Roots, stems, and leaves were separated and then dried at 70 °C to constant weight; thereafter, dry mass (g) of each was measured and subsequently combined to determine total seedling biomass. Subsamples of fresh leaves/needles were also used to estimate projected leaf area (cm²) based on total seedling leaf mass. For this, aspen leaves were scanned with a LI-3100 Area Meter (LI-Cor Inc., Lincoln, USA), while pine and spruce needles were scanned using a STD4800 scanner and analyzed using the WinSEEDLE™ software (Regent Instruments Inc. Quebec, Canada).

2.4. Statistical analysis

All analyses were conducted using R software, v 3.4.3, 64 bit (R Core Team, 2018a). Data for each question was averaged to the soil treatment cell level. Model residuals were tested for normality using the Shapiro-Wilk test from the R *stats* package (v 3.6.0; R Core Team, 2018b) and homogeneity of variance using Levene's test in the R *car* package (v 3.0-0; Fox and R Development Core Team, 2018); when data did not meet assumptions of normality or homogeneity, they were logarithmically transformed. Large operational scale studies like this generally have low replication, as a result we used an $\alpha = 0.1$ for all analyses to reduce the risk of a Type II error.

Differences in overall soil characteristics (i.e. initial soil nutrients, chemical characteristics) between all soil types were analyzed using a permutational ANOVA from the *lmPerm* package in R (v 2.1.0; Wheeler et al., 2016). One-way ANOVAs using linear mixed effects models (LMM) in the *nlme* package (v 3.1-131; Pinheiro et al., 2018) were used to compare the 2011 soil textures of each material, seasonal VWC, average seasonal and annual soil temperature, and the number of days with average soil temperature above 5 °C within each year (2013–2016), with soil treatment as the fixed effect and cell as the random effect.

To test the effects of coversoil material type, coversoil placement depth, and underlying subsoil treatments on annual tree growth from 2012 to 2016, a repeated measures ANOVA with LMMs was used, with soil treatment and year as the fixed effects and cell as the random effect. Comparisons of total tree height in 2016 were analyzed with fixed-effect linear models (LM) where soil treatment was set as the fixed effect. For the analysis of annual tree growth and total tree height in 2016 in response to coversoil depth, comparisons were individually made within each coversoil type. Total seedling biomass, leaf/stem/root mass and leaf area were exclusively compared between treatments 1 and 2 and analyzed with a one-way ANOVA using a full factorial design in the R *stats* package (v 3.6.0; R Core Team, 2018b). Interactions between the relative growth of all tree species and the soil treatments were analyzed with LMMs in a two-way ANOVA, where soil treatment and species were set as the fixed effects and cell as the random effect. All analyses, except for comparisons of relative growth among tree species, were run for each tree species separately.

When a significant main effect or an interaction were detected following any of the LMM analyses, LMMs were adjusted using the *lsmeans* function from the *lsmeans* package in R prior to running pair-wise comparisons (v 2.2-62; Lenth et al., 2018), and a Holm-Bonferroni adjusted $\alpha = 0.1$ using the contrast function from the R *car* package was used to conduct Fisher's LSD pair-wise comparisons (v 3.0-0; Fox and R Development Core Team, 2018). Note, least-squared means and adjusted standard errors of annual tree growth are reported in Figs. 3, 4, and 6, while original and unadjusted values are presented for all other data. For significant results found by a permutational ANOVA or fixed-effect LMs, the *LSD.test* function from the *agricolae* package was used to conduct pair-wise comparisons (v 1.2-8; de Mendiburu and R Development Core Team, 2017).

3. Results

3.1. Coversoil material type

Peat had a near-neutral pH and higher EC, SAR, OM, TOC, and TON than FFM (all $p < .1$; Table 1). While lower in plant available nitrate and sulfate, FFM was slightly acidic and contained more plant available phosphorus than Peat ($p < .1$; Table 1). FFM was coarse textured with 91.4% sand content, while mineral soil content in Peat was not measured (Table 1).

Peat (15 cm below the soil surface) had lower annual and seasonal soil temperatures than FFM, except for the 2016 growing season ($p < .01$ each year (2013–2015); Table 2). From 2013 to 2016, Peat was on average 1.9 °C colder; as a result, the Peat material had 8.4 fewer days each growing season where the daily average soil temperature was above 5 °C ($p < .05$ each year; Table 2). Volumetric water content (VWC) 15 cm below the soil surface was significantly higher in Peat for all years ($p < .001$ each year; Table 2). In the 2015 and 2016 growing seasons, water content at that depth never dropped below the wilting point for plants in the peat soil material (i.e. 25% VWC; Ojekanmi and Chang, 2014), whereas water content in the FFM repeatedly dropped below the wilting point of plants in sandy soils (i.e. 5% VWC; Saxton and Rawls, 2006) for extended periods of time (Fig. 2).

Aspen growth steadily increased in 2013 and 2014 on FFM, while on Peat aspen growth lagged and was not different between those years (Fig. 3a). In 2015, which had a notably dry growing season (Fig. 2; Section 2.1), growth of aspen in FFM decreased while it did not show this decrease in Peat; in the 2016 growing season, height growth of aspen did not recover on FFM while growth continued an upward trend on peat (Fig. 3a). Despite the decreased growth in FFM after 2015, aspen seedlings remained overall taller on the FFM compared to Peat in 2016 ($p < .05$; Fig. 3a). The aspen on the FFM also had higher stem and root mass in 2014 ($p < .1$ for both; Table 3). Pine growth significantly increased on FFM in 2013 and 2014 compared to Peat

Table 1

Summary of physical and chemical properties of the soil material types in 2012. Values represent adjusted means ($\pm$ SD) from the a) least-square means results for particle size data, and b) Fisher's LSD test results for all other data listed in the table. Letters indicate a significant difference among treatments ($\alpha \leq 0.1$).

		Peat	FFM	Subsoil Bm	Subsoil BC	Subsoil C
Sample Size		36	36	24	24	24
Cap Depth (cm)		10/30	10/20	30	130	100–130
Salvage (cm)		0–300	0–15	15–50	50–100	15–250
Particle Size Distribution (%) [†]	Sand	NM	91.4 ^B (2.8)	93.3 ^A (1.6)	93.8 ^A (2.6)	92.2 ^{AB} (5.8)
	Silt	NM	4.1 ^A (2.2)	2.4 ^{BC} (1.3)	1.3 ^C (1.2)	3.7 ^{AB} (4.1)
	Clay	NM	4.5 (0.9)	4.3 (0.7)	4.9 (1.5)	4.3 (1.8)
pH		7.47 ^A (0.19)	5.68 ^E (0.39)	6.04 ^D (0.47)	7.13 ^B (0.36)	6.76 ^C (0.76)
EC (dS·m ⁻¹)		1.25 ^A (0.32)	0.21 ^B (0.07)	0.17 ^B (0.06)	0.20 ^B (0.06)	0.19 ^B (0.16)
SAR		0.76 ^A (0.53)	0.20 ^C (0.04)	0.21 ^C (0.04)	0.30 ^B (0.27)	0.32 ^B (0.18)
OM (%)		31.2 ^A (11.7)	2.7 ^B (0.9)	NM	NM	NM
TOC(%)		15.6 ^A (5.8)	1.3 ^B (0.5)	NM	NM	NM
TON(%)		0.65 ^A (0.27)	0.04 ^B (0.01)	NM	NM	NM
Available Nutrients (mg per volume of soil)	NO ₃ ⁻	168.0 ^A (35.1)	38.7 ^C (1.2)	118.7 ^B (9.6)	102.0 ^B (0.0)	90.0 ^B (0.0)
	NH ₄ ⁺	16.8 ^B (1.2)	37.8 ^A (9.3)	14.3 ^B (1.6)	15.3 ^B (0.0)	16.7 ^B (1.7)
	P	87.0 ^D (2.1)	496.7 ^B (22.5)	1388.7 ^A (100.0)	588.6 ^B (39.9)	391.9 ^C (26.7)
	K	677.7 ^D (39.3)	808.2 ^C (53.8)	1345.7 ^A (60.0)	1266.5 ^{AB} (26.3)	1175.6 ^B (53.8)
	SO ₄ ⁻	7079.3 ^A (343.5)	104.6 ^B (9.2)	136.0 ^B (20.2)	133.9 ^B (17.3)	123.8 ^B (20.5)

Note: 'EC' refers to electrical conductivity, 'SAR' refers to sodium adsorption ratio, 'OM' refers to organic matter, 'TOC' refers to total organic carbon, 'TON' refers to total organic nitrogen, and 'NM' refers to not measured. [†] represents data that was logarithmically transformed prior to analysis.

($p < .05$ for both years; Fig. 3b). In 2015, pine improved growth over the previous years on both FFM and Peat materials, resulting in similar growth rates on both coversoil treatments that year (Fig. 3b). In 2016, growth decreased at a similar rate in both FFM and Peat (Fig. 3b); however, pine seedlings were still taller in 2016 on FFM than on Peat ($p < .05$; Fig. 3b). The taller pine trees in the FFM had higher total biomass, leaf, stem and root mass, and leaf area in 2014 ($p < .1$ for all; Table 3). Differences in spruce growth as a result of coversoil did not become apparent until 2016, where spruce on Peat grew more than on FFM ($p < .1$; Fig. 3c). Although there were clear differences in growth by the fifth growing season, total seedling height of spruce in 2016 was similar between the two coversoil materials (Fig. 3c). Spruce leaf area, total biomass, and root, stem, and leaf mass did not differ between the two coversoil materials in 2014 (Table 3).

When compared among species, relative growth of pine was the highest followed by aspen and spruce in both coversoil treatments (Fig. A2a). However, relative growth of pine and aspen differed between the two coversoils while it did not in spruce (coversoil $\times$ species interaction: $p < .1$; Fig. A2a). Root collar diameter responses to the coversoil treatments were similar to height growth in all species and are not presented.

3.2. Coversoil placement depth

Placement depth of Peat (10 cm (P10) vs. 30 cm (P30)) had an effect on soil temperature at 35 cm depth in all growing seasons, while placement depth did not influence soil temperature in FFM (10 cm (F10) vs. 20 cm (F20)) (2013–2016; Table 2). Soil temperature in P30 was on average 2 °C cooler each growing season than in P10 ($p < .05$ all years; Table 2), and P30 had 15 (2014) and 10 (2015) fewer days where average daily soil temperatures were above 5 °C than P10 ($p < .01$ in both years; Table 2). Despite being warmer than P30, P10 remained cooler compared to the two FFM placements in 2013 and 2014 ($p < .10$ in both years; Table 2). Volumetric soil water content (VWC) at a depth of 5 cm was significantly higher in P30 ($0.37\text{cm}^3\text{cm}^{-3}$) compared to P10 ($0.21\text{cm}^3\text{cm}^{-3}$), F10 ($0.05\text{cm}^3\text{cm}^{-3}$), and F20 ($0.06\text{cm}^3\text{cm}^{-3}$) each growing season ($p < .05$ for all). Similarly, VWC in P10 was significantly higher than both FFM placement depths each growing season ($p < .05$ (2013–2014, 2016)); however, VWC was similar among P10 ($0.16\text{cm}^3\text{cm}^{-3}$) and both FFM treatments (FFM10: $0.03\text{cm}^3\text{cm}^{-3}$; FFM20: $0.04\text{cm}^3\text{cm}^{-3}$) in the dry year of 2015.

Aspen growth did not differ between the placement depth

treatments for FFM in all years, while growth was lower on P30 compared to P10 in 2013 and 2014 ($p < .01$ in both years; Fig. 4a). By 2016, however, the total height of aspen trees did not differ between the Peat treatments (Fig. 4a). Pine growth was similar between the placement depths on the Peat and the FFM coversoils for all years (Fig. 4b), and as a result, total height of pine seedlings remained similar in the four treatments by 2016 (Fig. 4b). Spruce seedling growth in 2015 was slightly greater on P30 compared to P10, and by 2016 this difference was significant ($p < .1$; Fig. 4c). However, the total height of spruce trees measured in 2016 did not differ between the placement depths for FFM or Peat (Fig. 4c).

When comparing growth among species, pine had the highest relative growth compared to the other two species in both the P30 and P10 treatments (Fig. A2b). However, while aspen and pine positively responded to the thinner placement of Peat, spruce did not, resulting in a significant peat thickness by tree species interaction ($p < .1$; Fig. A2b). No interactions were found between placement depth and tree species in the FFM treatments. Root collar diameter responses to placement depth treatments were similar to height growth in all species and are not presented.

3.3. Subsoil material

All three subsoils were classified as sands; however, Subsoil BC had a lower silt content (1.3%) compared to Subsoils Bm and C (2.4% and 3.7%, respectively; $p < .05$ for BC vs. C only; Table 1). There were differences in pH between the subsoil materials, where pH was the highest in the BC (7.13) and lowest in the Bm (6.04) subsoil ($p < .1$; Table 1). The Subsoil Bm had higher plant available phosphorus compared to the other two subsoil materials ($p < .1$; Table 1).

Average seasonal VWC measured in the subsoil 45 cm below the soil surface was similar across all subsoil materials (Subsoil Bm: $0.097\text{cm}^3\text{cm}^{-3}$; Subsoil BC: $0.108\text{cm}^3\text{cm}^{-3}$; Subsoil C: $0.097\text{cm}^3\text{cm}^{-3}$). A subsoil treatment by year interaction was found in VWC 45 cm below the soil surface ($p < .05$) resulting from a drop in VWC in the BC material compared to the other subsoils in 2015 (data not shown). Volumetric soil water content measured in the rooting zone (15 cm depth) was similar among the three subsoils in the 2013 and 2014 growing seasons (Fig. 5). However, in 2015 when VWC decreased in all subsoil treatments, VWC in the Subsoil BC fell below the wilting point for plants in sand (i.e. 5% VWC; Saxton and Rawls, 2006), whereas Subsoils Bm and C remained above the threshold during these

Table 2
Mean ($\pm$ SD) annual and growing season soil temperature, number of days with an average daily soil temperature above 5 °C, and volumetric soil water content (VWC) at both 15 cm and 35 cm below the soil surface in FFM and Peat materials from 2013 to 2016 (n = 3). Letters indicate statistically significant differences between means within each response variable and each year (α = 0.1).

Placement Depth		2013				2014				2015				2016			
Sensor Depth		Peat		FFM		Peat		FFM		Peat		FFM		Peat		FFM	
		10 cm	30 cm	10 cm	20 cm	10 cm	20 cm	10 cm	30 cm	10 cm	20 cm	10 cm	30 cm	10 cm	20 cm	10 cm	30 cm
15 cm	Annual soil temperature (°C)	3.8 ^B (1.2)	4.0 ^B (0.7)	6.0 ^A (0.2)	6.0 ^A (0.3)	3.5 ^G (1.0)	2.7 ^G (0.4)	5.4 ^F (0.2)	5.1 ^F (0.2)	6.1 ^L (1.5)	4.3 ^M (0.5)	7.6 ^L (0.2)	6.1 ^L (0.4)	5.7 ^Y (0.6)	5.1 ^Y (0.4)	10.2 ^X (0.1)	6.3 ^Y (0.8)
	Seasonal soil temperature (°C)	11.3 ^B (0.5)	11.1 ^B (0.8)	13.3 ^A (1.1)	13.7 ^A (0.3)	10.5 ^G (1.1)	10.0 ^G (0.5)	12.1 ^F (0.9)	12.3 ^F (0.6)	12.2 ^{LM} (1.5)	11.2 ^M (0.3)	12.8 ^L (0.8)	12.9 ^L (1.1)	12.8 ^X (0.9)	12.4 ^X (0.1)	12.7 ^X (2.0)	13.5 ^X (1.3)
	# of days with average temp > 5 °C	140.7 ^B (2.5)	138.0 ^B (1.0)	149.3 ^A (2.1)	149.3 ^A (1.2)	140.0 ^{GH} (3.0)	138.3 ^H (1.5)	146.0 ^F (5.3)	144.7 ^{FG} (0.6)	149.0 ^{MN} (8.7)	144.3 ^N (0.6)	159.0 ^L (1.7)	156.0 ^{LM} (3.6)	159.0 ^X (3.0)	156.0 ^X (0.0)	139.3 ^Y (19.3)	160.3 ^X (4.9)
	VWC (cm ³ cm ⁻³)	0.09 ^B (0.02)	0.44 ^A (0.08)	0.09 ^B (0.01)	0.10 ^B (0.01)	0.09 ^G (0.02)	0.45 ^F (0.09)	0.08 ^G (0.01)	0.10 ^G (0.02)	0.08 ^M (0.01)	0.44 ^L (0.10)	0.07 ^M (0.01)	0.08 ^M (0.02)	0.08 ^Y (0.01)	0.43 ^X (0.1)	0.08 ^Y (0.004)	0.09 ^Y (0.02)
	Annual soil temperature (°C)	4.2 ^B (0.9)	4.3 ^B (0.8)	6.1 ^A (0.3)	6.1 ^A (0.4)	3.5 ^G (0.9)	2.4 ^H (0.6)	5.4 ^F (0.2)	5.2 ^F (0.4)	6.1 ^L (1.5)	4.0 ^M (0.5)	6.5 ^L (0.6)	6.0 ^L (0.2)	5.8 ^Y (0.4)	5.0 ^Z (0.1)	6.8 ^X (0.9)	6.4 ^{XY} (0.4)
35 cm	Seasonal soil temperature (°C)	10.7 ^B (0.9)	9.4 ^C (0.4)	12.0 ^A (0.7)	12.2 ^A (0.4)	9.6 ^G (0.8)	7.4 ^H (0.7)	10.8 ^F (0.8)	10.8 ^F (0.3)	11.4 ^L (1.1)	8.7 ^M (0.6)	11.5 ^L (0.6)	11.5 ^L (0.7)	12.2 ^X (0.8)	10.2 ^Y (0.6)	12.0 ^X (0.7)	12.4 ^X (0.9)
	# of days with average temp > 5 °C	126.0 ^B (22.6)	135.0 ^{AB} (4.0)	144.3 ^A (2.5)	141.3 ^{AB} (1.2)	136.0 ^F (2.0)	121.3 ^G (10.0)	140.3 ^F (1.5)	138.0 ^F (1.0)	143.3 ^M (1.5)	133 ^N (5.6)	154 ^L (2.6)	147 ^M (2.6)	155.0 ^X (1.0)	146.7 ^X (6.7)	143.3 ^X (27.3)	156 ^X (1.0)
	VWC (cm ³ cm ⁻³)	0.08 ^A (0.003)	0.09 ^A (0.01)	0.08 ^A (0.02)	0.10 ^A (0.01)	0.08 ^G (0.004)	0.09 ^G (0.01)	0.08 ^G (0.01)	0.10 ^F (0.01)	0.07 ^M (0.005)	0.09 ^{LM} (0.01)	0.08 ^{LM} (0.01)	0.09 ^L (0.01)	0.08 ^X (0.005)	0.08 ^X (0.01)	0.09 ^X (0.01)	0.10 ^X (0.01)
	Annual soil temperature (°C)	10.7 ^B (0.9)	9.4 ^C (0.4)	12.0 ^A (0.7)	12.2 ^A (0.4)	9.6 ^G (0.8)	7.4 ^H (0.7)	10.8 ^F (0.8)	10.8 ^F (0.3)	11.4 ^L (1.1)	8.7 ^M (0.6)	11.5 ^L (0.6)	11.5 ^L (0.7)	12.2 ^X (0.8)	10.2 ^Y (0.6)	12.0 ^X (0.7)	12.4 ^X (0.9)
	# of days with average temp > 5 °C	126.0 ^B (22.6)	135.0 ^{AB} (4.0)	144.3 ^A (2.5)	141.3 ^{AB} (1.2)	136.0 ^F (2.0)	121.3 ^G (10.0)	140.3 ^F (1.5)	138.0 ^F (1.0)	143.3 ^M (1.5)	133 ^N (5.6)	154 ^L (2.6)	147 ^M (2.6)	155.0 ^X (1.0)	146.7 ^X (6.7)	143.3 ^X (27.3)	156 ^X (1.0)

[†] Represents soil treatments with missing data during the growing season.

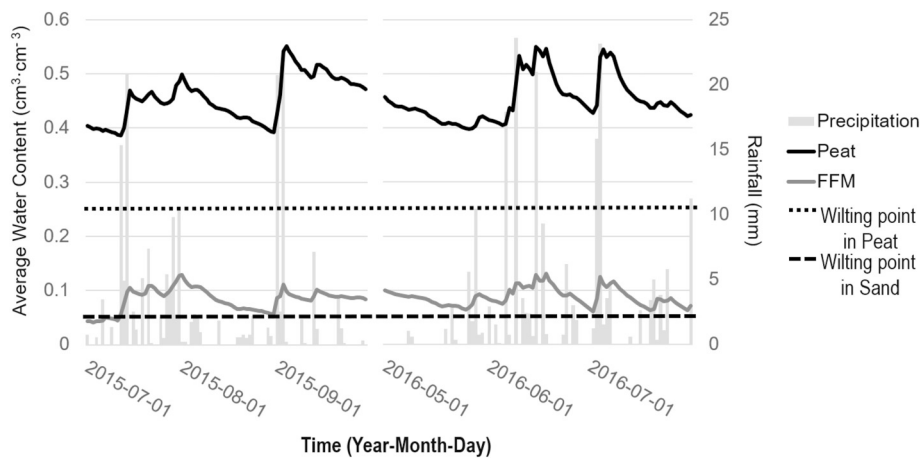


Fig. 2. Average daily volumetric water content for parts of the 2015 and 2016 growing seasons. Bars represent daily precipitation. Dashed lines represent an approximation of the wilting point for plants growing in peat ($0.25 \text{ cm}^3\cdot\text{cm}^{-3}$; 25% VWC) and sand ($0.05 \text{ cm}^3\cdot\text{cm}^{-3}$; 5% VWC) (Ojekanmi and Chang, 2014; Saxton and Rawls, 2006).

periods (subsoil $\times$ year interaction < 0.05 ; Fig. 5). By 2016, VWC remained below the wilting point in the Subsoil BC, while VWC increased in the Subsoil Bm and C treatments (Fig. 5).

Aspen growth was greater starting in 2014 on the Subsoil Bm and C treatments, while growth remained unchanged on the Subsoil BC treatment (Fig. 6a). In the dry year of 2015, growth was reduced in the

Subsoil BC treatment compared to Subsoil Bm and C (Fig. 6a), and in the following growing season, aspen growth on the Subsoil BC treatment decreased below the rates observed on Subsoil Bm ($p < .01$) and C ($p < .1$) (Fig. 6a). By 2016, aspen seedlings were tallest on the Subsoil Bm treatment and shortest on the treatments with Subsoil BC ($p < .005$) and C ($p < .01$; Fig. 6a). Despite the reduced height

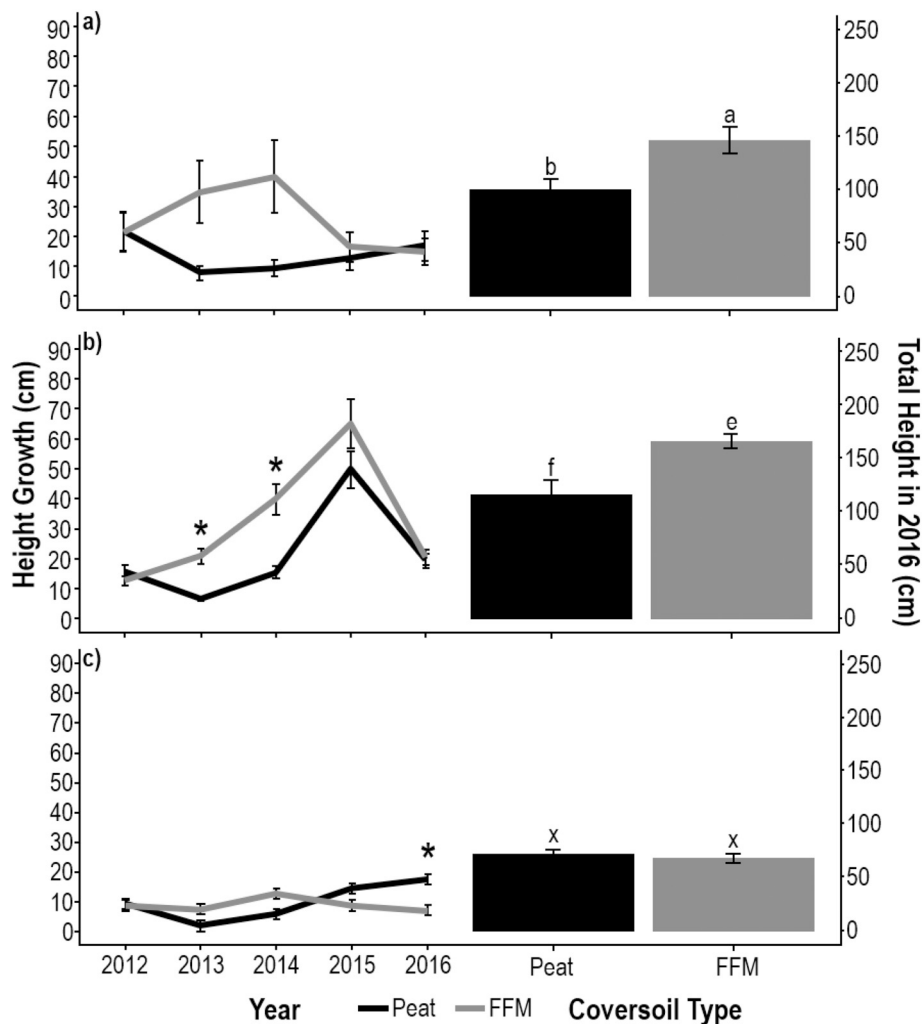


Fig. 3. Mean ($\pm$ SE) tree growth during the first five growing seasons (left panel) and average total tree height measured in 2016 (right panel) for trembling aspen (a), jack pine (b), and white spruce (c) growing in two coversoil material types. Asterisks (*) indicate significant differences between treatment growth means within a year, and different letters indicate significant differences between total heights ($\alpha \leq 0.1$, $n = 3$).

Table 3

Mean ($\pm$ SD) of leaf area (cm^2), total biomass, leaf, stem and root mass (g) in 2014 of aspen, pine and spruce on FFM and peat materials ($n = 3$). Different letters indicate statistically significant coversoil effects within each species ($\alpha \leq 0.1$).

Species	Topsoil material	(cm^2)	(g)	Leaf mass	Stem mass	Root mass
		Leaf area	Total biomass			
Aspen	FFM	5825 ^a (503)	338 ^a (13)	54.5 ^a (2.3)	122.6 ^a (10.2)	161.3 ^a (7)
	Peat	3099 ^a (2506)	180 ^a (130)	32.9 ^a (24.4)	61.4 ^b (48.2)	85.5 ^b (59.5)
Pine	FFM	6125 ^s (1091)	246 ^s (37)	123.4 ^s (21.5)	91.7 ^s (11)	31.3 ^s (6.9)
	Peat	2592 ^t (1636)	108 ^t (68)	55.3 ^t (34.4)	38 ^t (26.2)	15 ^t (8.1)
Spruce	FFM	2852 ^x (692)	121 ^x (30)	56.1 ^x (12.1)	44.8 ^x (11.8)	20.4 ^x (6.6)
	Peat	1924 ^x (1237)	100 ^x (58)	36.3 ^x (21.9)	43.2 ^x (25.2)	20.2 ^x (11)

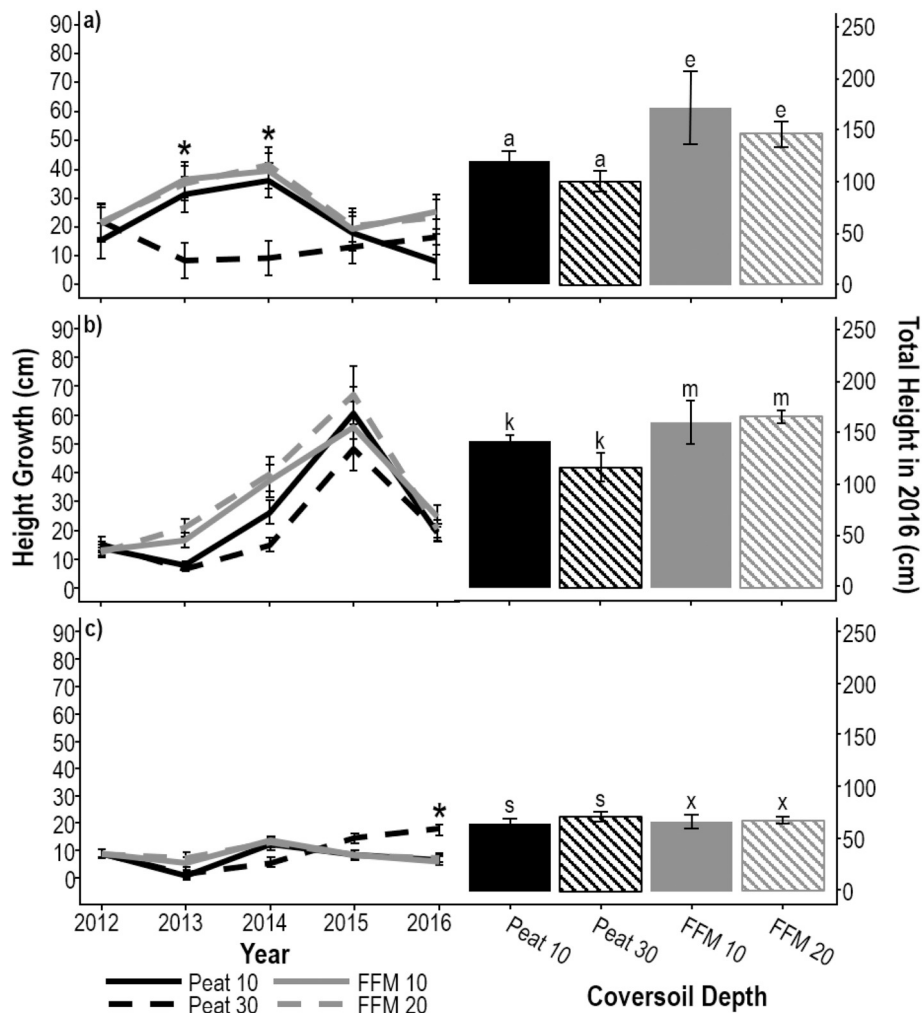


Fig. 4. Mean ($\pm$ SE) tree growth during the first five growing seasons (left panel) and average total tree height measured in 2016 (right panel) for trembling aspen (a), jack pine (b), and white spruce (c) in four coversoil placement depths. Asterisks (*) indicate significant differences between growth means of Peat or FFM depth treatments within a year, and different letters indicate significant differences between total heights within each coversoil type ($\alpha \leq 0.1$, $n = 3$).

growth of aspen on the Subsoil BC, RCD growth remained equal and consistent to seedlings grown in the other subsoils in 2014 (Bm: 19.4 mm, BC: 16.6 mm, and C: 16.3 mm) and 2016 (Bm: 26.6 mm, BC: 22.7 mm, and C: 22.3 mm). Pine growth rates started to differentiate in 2015 when seedlings grew slower on the Subsoil BC compared to the Subsoil Bm ($p < .005$) and Subsoil C ($p < .005$) (Fig. 6b); however, pine growth decreased similarly on all subsoil treatments during the 2016 growing season, resulting in no differences in growth that year (Fig. 6b). In 2016, total height of pine was greatest on treatments with Subsoils Bm and C compared to Subsoil BC (both $p < .05$; Fig. 6b). Root collar diameter of pine followed the same trends as height growth

and this data is not shown. Spruce growth only differed in 2014, where seedlings grew less on Subsoil C compared to the Subsoil BC ($p < .05$) (Fig. 6c). By 2016, total height of spruce was similar among the subsoil treatments (Fig. 6c). Root collar diameter of spruce followed the same trends as height growth and this data is not shown.

Relative growth differed among the three species, where pine had the greatest relative growth, followed by aspen, and spruce across all subsoil treatments (subsoil $\times$ tree species interaction: $p < .01$; Fig. A2c). Pine relative growth positively responded to the Subsoil Bm and C treatments, aspen responded more to the Subsoil Bm, and spruce did not respond to any subsoil treatment (Fig. A2c).

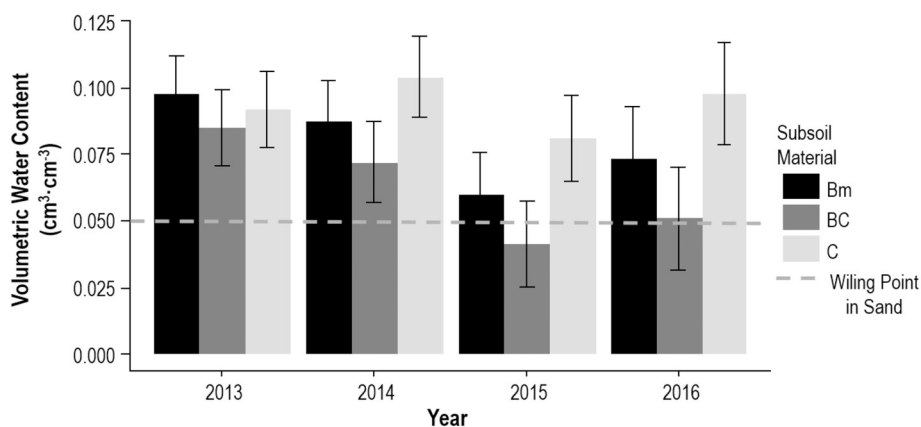


Fig. 5. Mean (± SE) growing season volumetric water content 15 cm below the soil surface in FFM treatments underlain by different subsoil materials in 2013, 2014, 2015, and 2016 ($n = 3$).

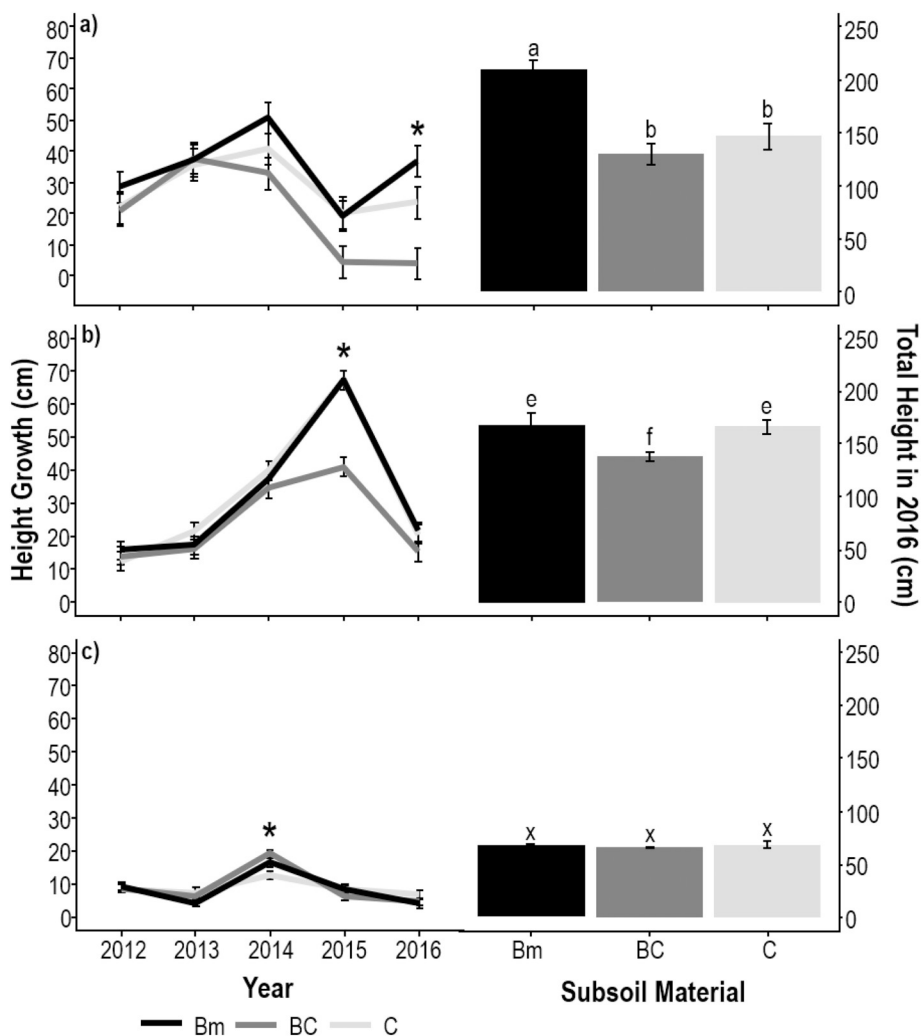


Fig. 6. Mean (± SE) tree growth during the first five growing seasons (left panel) and average total tree height measured in 2016 (right panel) for trembling aspen (a), jack pine (b), and white spruce (c) growing in three subsoil material types. Asterisks (*) indicate significant differences between treatment growth means within a year, and different letters indicate significant differences between total heights ($\alpha \leq 0.1$, $n = 3$).

4. Discussion

The type and amount of coversoil material placed at the soil surface had the greatest overall impact on early seedling growth (first five growing seasons) in our study. Aspen and pine responses were more pronounced on the two coversoil types and placement depths compared to spruce. Overall, both species grew significantly taller on the salvaged upland FFM compared to the lowland Peat coversoil. We identified potentially three factors and their interactions that could have affected resource availability and impacted seedling and species responses to the coversoil treatments. Differences could have been driven by the availability of less mobile nutrients such as phosphorus and by differences in the physical conditions of the coversoils (i.e. soil temperature and water holding capacity), which in turn affect resource availability (i.e. nutrients and water). Pioneer tree species, such as aspen and pine, rely heavily on the availability of essential nutrients, water supply, and warm soil temperatures to accommodate their early fast growth rates during establishment (Chapin III et al., 1986; Chapin III et al., 1983). The greater sensitivity of these species to soil conditions compared to white spruce partly explains their reduced growth rates observed on the Peat coversoil (Chapin III et al., 1986), with significantly lower levels of phosphorus and lower soil temperatures compared to the FFM. It also explains the decrease in aspen growth on the FFM when water availability became limiting during the dry growing conditions of 2015 compared to the wetter Peat coversoil. These strong responses are in contrast to white spruce, which responded comparatively weakly to the changing conditions. This was somewhat expected, as the relatively slower growing spruce is more tolerant to nutrient limitations and cooler root zone temperatures (Chapin III, 1977; Landhäusser et al., 2001, 2003; Zhang et al., 2013; Karst and Landhäusser, 2014).

Despite the phosphorus limitations, Peat had higher levels of plant available nitrogen, potassium, and sulfur, and SOM. While an abundance of SOM is beneficial for long-term nutrient cycling, the considerably higher SOM content in Peat compared to FFM had the greatest effect on root zone conditions. Peat can exhibit varying physical and chemical characteristics based on the source material and its degree of decomposition (Hayward and Clymo, 1982; Rezaeezhad et al., 2010); however, a common property associated with peat is its strong insulative abilities. In permafrost regions, the surface layers of peat will dry during the summer months, creating conditions of low thermal conductivity that increase temperature insulation with increasing soil depth; furthermore, peat can absorb large volumes of water that freeze during winter, increasing the thermal conductivity and subsequent penetration of sub-zero temperatures to greater depths (Nelson et al., 1985). In reclaimed mixtures of peat-mineral materials, Zhao and Si (2019) reported a decrease in thermal conductivity with an increase in the peat:mineral ratio, which corresponded with a subsequent decrease in soil temperature.

In our study, the insulative properties and high water content in the Peat layer delayed soil warming in the root zone each spring. Low soil temperatures have been shown to slow the metabolic activity and overall uptake of water and nutrients in tree roots, which inhibits important physiological processes such as root, leaf, and shoot growth (Landhäusser et al., 2001, 2003; Wan et al., 1999; Wolken et al., 2011). These low soil temperatures can also affect root-water dynamics in aspen seedlings by reducing water uptake, which directly impacts the uptake of nutrients and slows important functions such as stomatal conductance and net photosynthesis (Wan et al., 1999). In addition, cooler soil conditions resulted in delays at the start of each growing season, which likely led to the slower growth rates and shorter

seedlings, particularly in aspen and pine, by the fifth growing season. Root growth of most boreal tree species is considered to be limited below soil temperatures of 5 °C, although some species, such as white and black spruce, are more tolerant of low soil temperatures (Chapin III, 1977). Landhäusser et al. (2001) found minimal shoot and root growth in aspen when soil temperatures were below 5 °C compared to seedlings grown at 25 °C, while spruce showed no change in response to the same soil temperatures. Pine seedlings have shown sensitivity to low soil temperatures, resulting in reduced below- and aboveground growth rates (Karst and Landhäusser, 2014; Peng and Dang, 2003).

This relationship between soil temperature and seedling growth response was also apparent when comparing tree growth of aspen and pine between the shallow and deep Peat coversoil treatments, where seedlings negatively responded to the thicker 30 cm placement of Peat. The shallow Peat and FFM depth treatments had similar soil temperature and moisture conditions at 35 cm below the surface, where soil temperatures were warmer than the thick Peat treatment: these results demonstrate the soil temperature limitations introduced when placing SOM in thick layers at the soil surface. In a study of seedling survival following forest fire in the Canadian boreal, Greene et al. (2007) found a negative correlation between trembling aspen, jack pine, and black spruce with SOM thickness. Others suggested a layer > 20–30 cm of SOM will limit trembling aspen distribution, while black spruce can benefit from the insulative properties of thick SOM layers due to its shallow root system (Gewehr et al., 2014; Lafleur et al., 2015). The white spruce seedlings in our study reflect the relationship between black spruce and SOM layers, where seedlings had a slight positive response to the thicker placement of Peat, providing additional support to the ability of this species to maintain growth in cooler soil temperatures (Landhäusser et al., 2003; Wolken et al., 2011).

While the high SOM and associated insulative properties of Peat created temperature limitations, there is a potential trade-off with the ability of peat to store and supply water. This trade-off became evident during the dry conditions of 2015–2016, where trees grew noticeably less in the thin Peat (10 cm) and the FFM coversoil treatments compared to the thicker Peat (30 cm) treatment. When combined with the soil water content data, water availability from the upper surface layers where most roots are located suggest that the thin Peat and the FFM coversoil treatments did not hold enough water to buffer against the dry growing conditions of 2015 (Section 3.2). Water availability in the FFM was low as VWC reached the permanent wilting point (PWP) of sandy soil textures for extended periods during the 2015 and 2016 growing period; alternatively, VWC in Peat remained above the PWP for this material type and provided seedlings with enough water to maintain growth. Limited water availability negatively affects photosynthetic processes and reduces the uptake of nutrients that are essential for physiological functions (Fageria, 2013; Kreuzwieser and Gessler, 2010); thus, it is important to consider the cover soil materials and their water storage capabilities during dry growing seasons. These findings indicate the important role that Peat materials can have in soil water and plant dynamics on upland forest reclamation, especially if there is a prolonged period of drought conditions or if dry growing conditions become more frequent with a changing global climate (Kreuzwieser and Gessler, 2010; Krishnan et al., 2006).

Interestingly, pine seedlings grew more during the dry season of 2015 compared to the other years measured. This greater growth response in pine may have partially been influenced by the conditions of the previous growing season, where the area received above average precipitation during the time when the new buds for 2015 were formed (Burns and Honkala, 1991). Therefore, an early spring in 2015 coupled

with the effects of a wet 2014 season may have been the cause for greater pine growth in 2015; however, the effects of the dry 2015 growing season were noticeable in the growing season following the dry year when pine growth decreased significantly.

While the type of coversoil had a substantial effect on the growth response of our seedlings, the type of subsoil material placed below the coversoil also had a measurable effect on early seedling establishment and growth. As root growth of seedlings on the Peat coversoil was limited in the first five growing seasons, potential responses to the different subsoil treatments were likely constrained and obscured by the strong coversoil effect. Thus, we used the FFM capped subsoil treatments to explore these responses to different subsoils as the seedlings had much larger tree root systems that had accessed the underlying subsoil materials later in the study period. Some of the differences we detected among the different subsoil types appear to be driven by the availability of mineral nutrients. An increased proportion of the lower C horizon in the subsoil B layer likely diluted the nutrients that were concentrated in the weathered B horizon (Ojekanmi and Chang, 2014), which is noticeable when comparing the higher concentrations of phosphorus in the Subsoil Bm layer to the homogenous BC and C subsoils. Despite the additional supply of phosphorus in this subsoil, aspen was the only species that responded positively to the Subsoil Bm treatment. Aspen is known for investing more resources into root systems during the early years of establishment, growing expansive and deep roots for rapid access to available nutrients in the deeper soil layers compared to pine and spruce (Strong and La Roi, 1983; Strong and La Roi, 1985). However, the greater root growth is also associated with a greater demand for essential nutrients and with greater phosphorus availability in the weathered layer of the Bm treatment, aspen would have had access to this nutrient, provided other essential nutrients were not becoming limited as their roots expanded deeper into the soil profile (Chapin III et al., 1986; Chapin III et al., 1983; Pinno et al., 2012).

Water availability was another potential driver that could have affected the growth response of the seedlings to the different subsoils. Although the three subsoils were classified as sands, average daily water content during the very dry 2015-growing season was significantly higher at 15 cm depth in soils reconstructed with Subsoil C compared to Subsoil Bm or BC. Although the subsoil materials had similar textures and bulk densities, there were slight but significant differences in silt content that corresponded with the average water content each growing season. The Subsoil C treatment had the highest silt and water content and Subsoil BC had the lowest. Studies on coarse-textured boreal soils reported that even small differences in silt content (1–3%) between the subsoil horizons can make a large difference in the water holding ability of sandy soils (Gale and Grigal, 1987; Huang et al., 2013; Jung et al., 2014; Zettl et al., 2011). The slightly higher silt contents in the treatments with Subsoil Bm or Subsoil C may have held more water in the root zone by slowing drainage past the FFM and subsoil interface, which could explain the higher water content measured 15 cm below the soil surface compared to 45 cm (Section 3.3).

Large controlled reclamation field studies like the ASCS are rare and provide a unique opportunity to test a range of ecological questions that are difficult to conduct in natural environments. In this study, mixing a coarse mineral soil material (particularly the surface Bm material) with the peat could have potentially provided an opportunity to develop a surface soil material that balances nutrient availability with higher water holding capacity for trees during dry years, as well as warmer soils and faster spring warming compared to a peat-only coversoil.

Similarly, the coarse FFM could have benefitted by amending it with Peat to improve the water holding capacity of this otherwise suitable coversoil during dry growing conditions and providing more suitable seedbeds for the upland propagules that were retained in the salvaged FFM (Jones and Landhäusser, 2018). While the root systems of most tree species were just starting to explore the deeper soil layers in this study, there were indications that the type of subsoil material will impact the growth of trees and possibly their resilience to stresses such as drought in the long-term. The dry conditions observed in the latter years of this study demonstrate how textural heterogeneity may be important for improving water availability in coarse-textured subsoils. Furthermore, the placement of selectively salvaged and weathered subsoils may provide additional nutrients as tree roots continue to grow deeper into the reconstructed soil profile. The knowledge gained from this study will be useful in the development of operational criteria for reconstructing soil covers used in boreal forest reclamation, while informing the impacts of soil covers on afforestation success and furthering the development of functional and sustainable forest ecosystems.

Declaration of Competing Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

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Appendix A. Appendix



Fig. A1. Map of the Aurora Soil Capping Study (ASCS). The soil layering treatment is designated by a circle in each cell, and tree plots are designated as squares with their planting treatment identified within. ‘A’ refers to trembling aspen, ‘P’ refers to jack pine, and ‘S’ refers to white spruce.

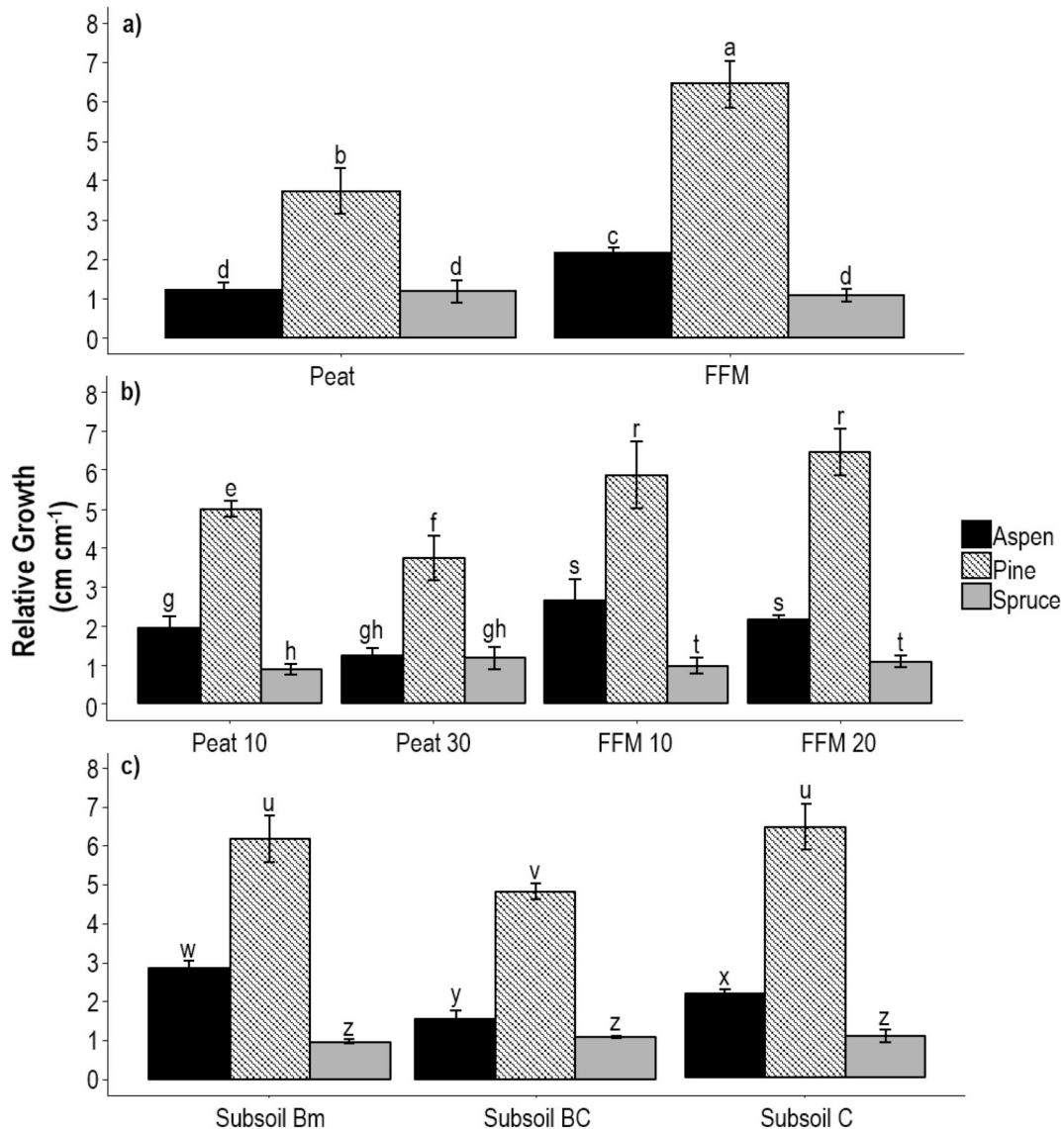


Fig. A2. Mean ($\pm$ SE) tree growth from 2012 to 2016 relative to initial 2012 height for trembling aspen, jack pine, and white spruce in different: a) coversoil material types, b) depths of Peat and FFM, and c) subsoil material types. Letters indicate significant differences between tree species growth means and soil treatment ($\alpha \leq 0.1$, $n = 3$). Relative growth in b) was analyzed separately within Peat and FFM coversoils.

Table A1

Average daily temperatures (beginning 10 days after first daily average temperature > 5 °C) and total growing season precipitation during the first five growing seasons.

	2012	2013	2014	2015	2016
Average Daily Temperature (°C)	15.5	16.9	15.1	14.5	15.2
Growing Season Precipitation (mm)	253.2	266.9	315.4	209.1	342.8

Table A2

Mean (± SD) initial seedling characteristics at out planting in 2012 (n = 20).

Species	Height (cm)	Shoot mass (g)	Root mass (g)	RSR	Plug height (cm)	Plug diameter (cm)	Plug volume (ml)
Aspen	30 (8.6)	0.7 (0.4)	2.5 (1.2)	3.4	15	6	340
Pine	18.2 (2.8)	2.2 (0.7)	1.3 (0.5)	0.6	12	4	125
Spruce	29 (5.5)	4.5 (1.6)	2.9 (1.4)	0.7	15	6	340

RSR = Root shoot ratio.

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